

Telecom Customer Churn Analytics Report

Project Domain: Telecom Analytics

Project Type: End-to-End Data Analytics Project

Technologies Used:

Python | Machine Learning | SQL | Excel | Power BI

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Technical Stack

Programming Language

- **Python**
Used for data cleaning, exploratory data analysis (EDA), feature engineering, and machine learning model development.

Databases

- **SQL (MySQL)**
Used for structured data storage, querying churn-related patterns, and validating analytical results through aggregations and joins.

Visualization Tools

- **Power BI**
Used to design an interactive, multi-page dashboard with KPIs, slicers, and predictive insights using DAX and Power Query.
- **Microsoft Excel**
Used for pivot table analysis, churn validation, KPI summaries, and intermediate business-level analysis.

Libraries & Frameworks

- **Pandas** – Data manipulation and transformation
- **NumPy** – Numerical computations
- **Matplotlib & Seaborn** – Data visualization during EDA
- **Scikit-learn** – Machine learning model development and evaluation
- **XGBoost** – Advanced gradient boosting model for churn prediction

Machine Learning Models Used

- **Logistic Regression** – Baseline classification model
- **Random Forest Classifier** – Ensemble-based churn prediction model
- **XGBoost Classifier** – Optimized model for improved churn prediction performance

Project Overview

Customer churn is a major challenge in the telecom industry, directly affecting revenue and customer retention. This project, ChurnX Insights — Telecom Customer Analytics Report, focuses on analyzing customer behavior and predicting churn using an end-to-end data analytics approach. Python is used for data cleaning, exploratory analysis, and machine learning modeling, while SQL and Excel support data validation and business analysis. Power BI is used to build an interactive, multi-page dashboard for decision-making. The project combines descriptive and predictive analytics to identify churn patterns and high-risk customers. The final output delivers actionable insights that help businesses design effective retention strategies.

Business Problem Statement

Telecom companies face high customer churn due to competitive pricing, service quality issues, and contract flexibility. Identifying customers likely to churn before they leave is a critical business challenge. Traditional reports provide historical insights but fail to predict future churn risks. This project addresses the need to analyze churn drivers and predict customer churn using data-driven techniques. By applying machine learning and advanced analytics, the project helps businesses proactively target high-risk customers. The goal is to reduce churn, improve customer retention, and support informed business decisions.

Project Objectives

- To analyze customer behavior and identify key factors contributing to customer churn in the telecom industry.
- To perform exploratory data analysis (EDA) to uncover churn patterns across contracts, services, tenure, and payment methods.
- To build and evaluate machine learning models for predicting customer churn probability.
- To classify customers into risk categories and identify high-risk churn segments.
- To validate analytical insights using SQL queries and Excel pivot-based analysis.
- To design an interactive, multi-page Power BI dashboard for business decision support and stakeholder insights.

Dataset Description

The dataset used in this project is a publicly available **Telecom Customer Churn dataset**, commonly used for churn analysis and predictive modeling. It contains customer-level information related to demographics, service subscriptions, contract details, and billing attributes.

The dataset consists of approximately **7,000 customer records** with **multiple categorical and numerical features**. The target variable is **Churn**, which indicates whether a customer has discontinued the telecom service. Key attributes include tenure, contract type, internet service, payment method, monthly charges, total charges, and service usage indicators. This dataset provides a comprehensive view of customer behavior and is well-suited for analyzing churn patterns and building predictive models.

Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand customer behavior and identify patterns associated with churn. Initial analysis focused on the overall churn distribution, revealing a significant portion of customers discontinuing services.

Key analyses included churn trends across **contract types, tenure groups, internet services, and payment methods**. Customers on month-to-month contracts and those using fiber optic internet exhibited higher churn rates. Short-tenure customers were more likely to churn compared to long-term subscribers.

These insights helped identify critical churn drivers and guided feature selection for machine learning models, as well as the design of business-focused dashboards.

Machine Learning Methodology

The machine learning process began with splitting the dataset into training and testing sets to evaluate model performance effectively. Categorical variables were encoded, and numerical features were scaled where required to ensure model compatibility.

Multiple classification models were implemented to predict customer churn, including **Logistic Regression, Random Forest, and XGBoost**. Logistic Regression served as a baseline model, while ensemble methods were used to capture complex, non-linear relationships in customer behavior.

Model training and tuning were performed using appropriate evaluation metrics, ensuring a balanced comparison of performance across different algorithms.

SQL Analysis

SQL was used to perform structured analysis and validate churn insights derived from Python and machine learning workflows. The cleaned dataset was stored in a relational database, enabling efficient querying and aggregation of customer data.

Key SQL queries were executed to analyze churn patterns across contract types, internet services, payment methods, and tenure groups. These queries helped identify high-risk segments and ensured consistency between analytical results obtained from different tools.

The SQL analysis served as a validation layer, strengthening the reliability of insights and supporting cross-platform analytical accuracy.

Excel Analysis & Validation

Microsoft Excel was used to validate analytical findings and present churn insights in a business-friendly format. Pivot tables were created to analyze churn distribution across contract types, internet services, payment methods, tenure groups, gender, and senior citizen status.

Charts and KPI summaries were developed to visually validate trends identified through SQL and Python analysis. This step ensured consistency of churn patterns across multiple platforms and improved confidence in the results.

Excel served as an intermediate validation layer, bridging technical analysis and business interpretation before final dashboard development in Power BI.

Power BI Dashboard Overview

Power BI was used to design an interactive and visually intuitive dashboard that presents churn insights in a business-oriented manner. The dashboard integrates cleaned data, engineered features, and machine learning outputs to support decision-making at multiple levels.

A dark-themed user interface was implemented to enhance visual clarity and user experience. DAX measures were created to calculate KPIs such as churn rate, average tenure, average monthly charges, and high-risk customer percentage. Interactive slicers allow users to filter insights by demographics, services, contracts, and payment methods.

The dashboard serves as a centralized analytics solution, transforming complex data into actionable business insights.

Power BI Dashboard

The Power BI dashboard is designed as a **five-page interactive report**, with each page focusing on a specific business perspective of customer churn.

1. Executive Summary

Provides a high-level overview of key KPIs such as total customers, churn rate, average tenure, and high-risk customers. Interactive slicers enable quick filtering, allowing leadership to understand churn trends within seconds.

2.Customer Demographics Analysis

Analyzes churn patterns across gender, senior citizen status, partner and dependent relationships, and tenure groups. This page helps identify demographic segments that are more prone to churn.

3.Services & Usage Behavior

Focuses on churn trends based on subscribed services, including internet type, tech support, streaming services, and security options. It highlights service combinations that significantly influence churn risk.

4.Billing & Payment Insights

Examines the impact of payment methods, monthly charges, total charges, and revenue categories on churn. This page provides insight into billing-related churn drivers.

5.Churn Risk Prediction (ML Insights)

Displays machine learning-based churn probability analysis, including high-risk customer percentage and predictive insights. This page bridges analytics and business action by identifying customers requiring immediate retention efforts.

Key Insights & Findings

- Customers on month-to-month contracts show the highest churn rate, indicating low customer commitment.
- Fiber optic internet users have a higher churn tendency compared to DSL and non-internet users.
- Customers with shorter tenure are more likely to churn than long-term customers.
- Electronic check payment method is associated with higher churn compared to automatic payment options.
- Customers with higher monthly charges tend to exhibit increased churn risk.
- Machine learning models identified a significant portion of customers as high-risk, enabling proactive churn prevention.

Business Recommendations

- Encourage customers to shift from month-to-month contracts to long-term plans by offering discounts and loyalty benefits.
- Improve service quality and pricing strategies for fiber optic internet users to reduce dissatisfaction and churn.
- Implement early engagement programs for new customers to reduce churn during the initial tenure period.
- Promote automatic payment methods by offering incentives to customers using electronic check payments.
- Introduce personalized retention offers for customers identified as high-risk through machine learning predictions.
- Use churn probability insights to design targeted marketing and customer support strategies.

Limitations of the Project

- The dataset used in this project is a historical snapshot and does not capture real-time customer behavior.
- Some customer behavior factors such as customer satisfaction, network quality, and competitor offer are not included in the dataset.
- The machine learning models are limited by the quality and scope of available features.
- Class imbalance in churn data may impact prediction performance for churned customers.

- The project assumes consistent customer behavior patterns, which may change over time.

Future Scope

- Integrate real-time customer data to enable live churn monitoring and alerts.
- Enhance prediction accuracy by experimenting with advanced machine learning and deep learning models.
- Incorporate additional features such as customer feedback, network performance, and usage patterns.
- Automate churn risk alerts and integrate insights with CRM systems for proactive customer retention.
- Deploy the churn prediction model as a production-ready solution for continuous business use.

Conclusion

This project successfully delivered an end-to-end telecom customer churn analytics solution by integrating data analysis, machine learning, and business intelligence tools. Through detailed exploratory analysis and predictive modeling, key churn drivers and high-risk customer segments were identified. The interactive Power BI dashboard transformed complex analytical results into actionable business insights. Overall, the project demonstrates strong analytical, technical, and visualization skills while providing meaningful value for customer retention and decision-making.